

Statistical and Machine Learning

Evaluation of model performance

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Exercise 1 – Confusion Matrix and Basic Metrics

A binary classifier is evaluated on 200 patients (cancer vs healthy). The results are: $TP = 40$ (true positives, i.e., a person having cancer was predicted as such), $FP = 10$ (false positives), $FN = 20$ (false negatives), $TN = 130$ (true negatives).

1. Fill in the confusion matrix for this classifier.

Model answer:

	Predicted Positive	Predicted Negative
Actually Positive		
Actually Negative		

Total = ?

2. Compute the **accuracy**, **sensitivity**, **specificity**, and **precision**.
3. Suppose this classifier is used to screen for cancer. Which metric is most important – sensitivity or specificity – and why? In other words, which metric should we maximise?
4. A second classifier has $TP = 55$, $FP = 30$, $FN = 5$, $TN = 110$. Compute its accuracy. Explain why accuracy alone is insufficient to compare the two classifiers in this context.

Exercise 2 – F1 Score and the Precision-Recall Trade-off

1. Using the counts from Exercise 1 ($TP = 40$, $FP = 10$, $FN = 20$, $TN = 130$), compute the **F1 score**.
2. In general, lowering the classification threshold increases recall but decreases precision. Explain intuitively why this is the case.
3. Give one real-world scenario where you would prefer **high recall** over high precision, and one where you would prefer **high precision** over high recall.

Exercise 3 – ROC Curve and AUC

1. Explain what the ROC curve plots on each axis, and what the role of the threshold t is.
2. What does an AUC of 0.5 correspond to, and what does an AUC of 1.0 correspond to? What is the AUC of a classifier that always predicts the majority class?

- Two models A and B have $AUC = 0.85$ and $AUC = 0.78$ respectively. Model A is preferred overall, but a doctor requires that the false positive rate never exceed 5%. Can AUC alone determine which model to use? Explain.
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Exercise 4 – Regression Loss Functions

- Write down the L_1 , L_2 , and Huber loss functions.
 - A dataset contains 100 observations with small residuals, plus 5 highly influential observations. Sketch (or describe) what happens to the fitted regression line when you use L_2 vs L_1 loss, and explain the difference.
 - The Huber loss has a tuning parameter δ . What happens to the Huber loss as $\delta \rightarrow \infty$? What happens as $\delta \rightarrow 0$?
 - Give one practical reason why L_2 is more commonly used than L_1 despite L_1 being more robust to influential observations.
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Exercise 5 – Cross-Validation

- Describe the K -fold cross-validation procedure in your own words. What is the purpose of holding out one fold at a time?
 - Write down the cross-validation estimate of prediction error $CV_{(K)}$.
 - Consider $K = n$ (leave-one-out CV) and $K = 2$. Describe one advantage and one disadvantage of each.
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Exercise 6 – Bias–Variance Trade-off and Double Descent

- State the bias-variance decomposition of the MSE. Define each term.
 - A polynomial regression model is fitted with degrees $d = 1, 2, \dots, 15$. Sketch (or describe) the expected shape of the training error and test error curves as d increases. What is the name of the phenomenon at high d ?
 - Describe the **double descent** phenomenon. At what point does the second descent begin, and what drives it?
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Exercise 7 – Bootstrap

- Describe the bootstrap procedure for estimating the standard error of an estimator $\hat{\theta}$. What is the key difference from classic repeated sampling?
- In the portfolio example, $\bar{\hat{\alpha}} = 0.666$ and $\hat{\sigma}(\hat{\alpha}) = 0.076$. Interpret these two numbers.
- Name one assumption under which bootstrap resampling faithfully mimics population sampling, and one situation where it might fail.